

Maintain Environmental Health and Safety

November/December 2025

SECTION A (40 MARKS)

Attempt ALL the questions in this section.

1. Developing a near miss plan helps in preventing occurrence of accidents in an industry. Name THREE examples of near miss. (3 Marks)
2. Housekeeping refers to the management and maintenance of a physical space to keep it clean, orderly, and safe. List FOUR reasons for maintaining housekeeping in a workplace. (4 Marks)
3. Risk assessment involves evaluating potential hazards and determining their impact on systems or operations. Within this framework, distinguish between Event Tree Analysis and Fault Tree Analysis. (4 Marks)
4. A well-stocked first aid kit is essential for treating minor injuries and stabilizing a person before professional medical help arrives. Give FOUR components of a first Aid kit. (4 Marks)
5. Emergency evacuation is a critical aspect of industrial safety. State FOUR key components of an effective emergency response plan. (4 Marks)
6. As part of environmental regulations, industries must treat their waste prior to disposal. Identify FOUR methods commonly used for industrial waste disposal. (4 Marks)
7. Environmental Health and Safety (EHS) is the discipline and practices focused on protecting the environment, ensuring the health and safety of people. Define the following terms as used in EHS. (4 Marks)
 - a) Evacuation.
 - b) Safety plan.
8. It is a legal requirement for industries and commercial buildings to place and maintain fire signage's in place. List FOUR fire signage's that are found in a Chemical Engineering workshop. (4 Marks)
9. Workplace preparedness for emergencies requires appropriate safety equipment. Name FIVE types of safety equipment typically found in industrial setting. (5 Marks)

10. Safety standards are set by international bodies to ensure harmonization and uniformity of standards. Outline TWO safety standards commonly followed in maintaining workplace safety and environmental health. (4 Marks) **SECTION B (60 MARKS)**

Attempt ANY THREE (3) questions in this section

11.

- a) An emergency drill is a simulated exercise conducted to practice the correct response to various emergency situations. Describe FIVE benefits of conducting emergency drills. (10 Marks)
- b) An evacuation plan is a detailed and organized procedure designed to ensure the safe and efficient removal of people from a building or area during an emergency. Explain FIVE components of evacuation plan. (10 Marks)

12.

- a) The process of recognizing any source or situation with the potential to cause harm is the first step in risk assessment. Describe FIVE benefits of identifying near misses and incidences according to environmental health and safety standard. (10 Marks)
- b) Water pollution occurs when harmful substances contaminate water bodies making it unfit for human and aquatic life. Explain FIVE major water pollutants. (10 Marks)

13.

- a) ISO standards are internationally recognized guidelines that provide a framework for organizations to improve their management practices and processes. ISO 14000 family of standards was developed to deal with Environmental management. Describe SIX benefits of ISO 1400 standard. (12 Marks)
- b) Fire is classified into different classes depending on the cause and type of material burning. Explain FOUR classes of fire. (8 Marks)

14.

- a) After recruiting new employees in a manufacturing industry, Environmental Safety Officers are responsible for conducting safety training. Describe FOUR essential elements that should be considered when carrying out these training sessions. (8 Marks)

- b) To maintain a safe working environment, industries must implement systems to monitor occupational hazards that could potentially harm employees' health or safety. Explain FOUR monitoring systems used to identify and manage these hazards. (8 Marks)
- c) Personal Protective Equipment (PPE) is a critical component of workplace safety programs and should be used in conjunction with other safety measures. Name benefits of PPE in the workplace. (4 Marks)